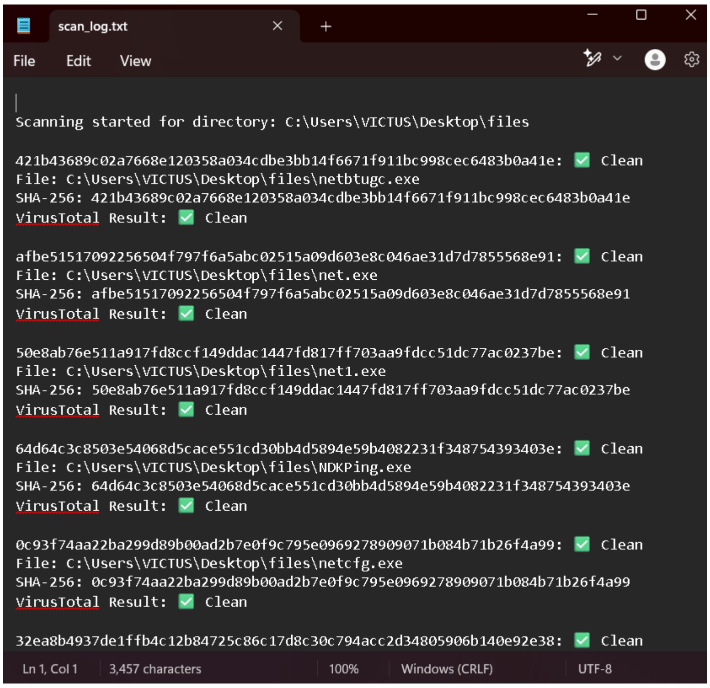
Continuing Minor Project (Malware Detection Tool)

Implementing basic scanning functionality in your Python-based malware detection tool.
Trying to detect malicious files based on hash signatures.

1.Added VirusTotal Results to Output



```
scan_log.txt
File  Edit  View

Scanning started for directory: C:\Users\VICTUS\Desktop\files

421b43689c02a7668e120358a034cdbe3bb14f6671f911bc998cec6483b0a41e: ✅ Clean
File: C:\Users\VICTUS\Desktop\files\netbtugc.exe
SHA-256: 421b43689c02a7668e120358a034cdbe3bb14f6671f911bc998cec6483b0a41e
VirusTotal Result: ✅ Clean

afbe51517092256504f797f6a5abc02515a09d603e8c046ae31d7d7855568e91: ✅ Clean
File: C:\Users\VICTUS\Desktop\files\net.exe
SHA-256: afbe51517092256504f797f6a5abc02515a09d603e8c046ae31d7d7855568e91
VirusTotal Result: ✅ Clean

50e8ab76e511a917fd8ccf149ddac1447fd817ff703aa9fdcc51dc77ac0237be: ✅ Clean
File: C:\Users\VICTUS\Desktop\files\net1.exe
SHA-256: 50e8ab76e511a917fd8ccf149ddac1447fd817ff703aa9fdcc51dc77ac0237be
VirusTotal Result: ✅ Clean

64d64c3c8503e54068d5cace551cd30bb4d5894e59b4082231f348754393403e: ✅ Clean
File: C:\Users\VICTUS\Desktop\files\NDKPing.exe
SHA-256: 64d64c3c8503e54068d5cace551cd30bb4d5894e59b4082231f348754393403e
VirusTotal Result: ✅ Clean

0c93f74aa22ba299d89b00ad2b7e0f9c795e0969278909071b084b71b26f4a99: ✅ Clean
File: C:\Users\VICTUS\Desktop\files\netcfg.exe
SHA-256: 0c93f74aa22ba299d89b00ad2b7e0f9c795e0969278909071b084b71b26f4a99
VirusTotal Result: ✅ Clean

32ea8b4937de1ffb4c12b84725c86c17d8c30c794acc2d34805906b140e92e38: ✅ Clean
Ln 1, Col 1   3,457 characters   100%   Windows (CRLF)   UTF-8
```

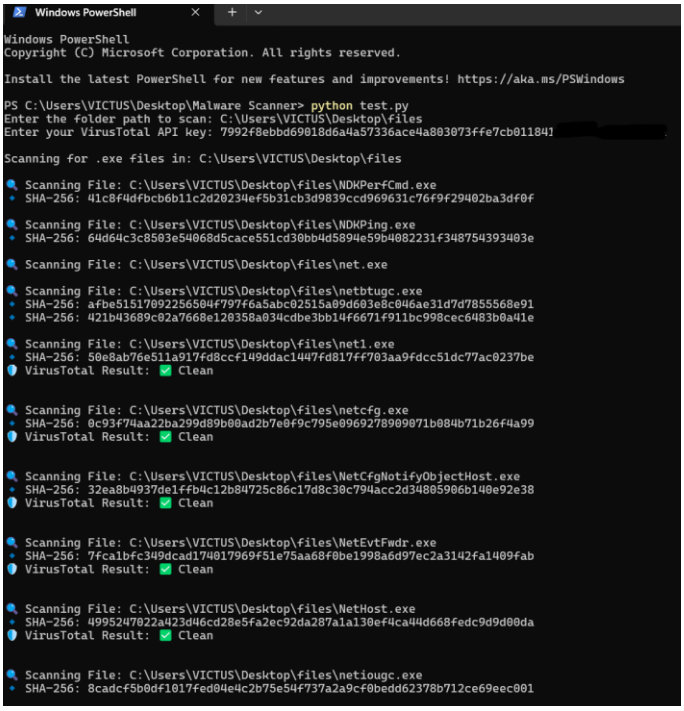
- **Before fix :** The script **only displayed file names and SHA-256 hashes**—it didn't indicate whether a file was **malicious or clean.**
- **After Fix:** Now, the script clearly shows if a file is:

1.Clean

2.Malicious (with detection count)

3. Not Found in VirusTotal

2. Parallel Processing for Faster Scanning



```
Windows PowerShell

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\VICTUS\Desktop\Malware Scanner> python test.py
Enter the folder path to scan: C:\Users\VICTUS\Desktop\files
Enter your VirusTotal API key: 7992f8ebbd69018d6a4a57336ace4a803073ffe7cb01184[redacted]

Scanning for .exe files in: C:\Users\VICTUS\Desktop\files

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\NDKPerfCmd.exe
🔹 SHA-256: 41c8f4dfbcb6b11c2d20234ef5b31cb3d9839ccd969631c76f9f29402ba3df0f

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\NDKPing.exe
🔹 SHA-256: 64d64c3c8503e54068d5cace551cd30bb4d5894e59b4082231f348754393403e

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\net.exe

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\netbtugc.exe
🔹 SHA-256: afbe51517092256504f797f6a5abc02515a09d603e8c046ae31d7d7855568e91
🔹 SHA-256: 421b43689c02a7668e120358a034cdbe3bb14f6671f911bc998cec6483b0a41e

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\net1.exe
🔹 SHA-256: 50e8ab76e511a917fd8ccf149ddac1447fd817ff703aa9fdcc51dc77ac0237be
🛡️ VirusTotal Result: ✅ Clean

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\netcfg.exe
🔹 SHA-256: 0c93f74aa22ba299d89b00ad2b7e0f9c795e0969278909071b084b71b26f4a99
🛡️ VirusTotal Result: ✅ Clean

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\NetCfgNotifyObjectHost.exe
🔹 SHA-256: 32ea8b4937de1ffb4c12b84725c86c17d8c30c794acc2d34805906b140e92e38
🛡️ VirusTotal Result: ✅ Clean

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\NetEvtFwdr.exe
🔹 SHA-256: 7fca1bfc349dcad174017969f51e75aa68f0be1998a6d97ec2a3142fa1409fab
🛡️ VirusTotal Result: ✅ Clean

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\NetHost.exe
🔹 SHA-256: 4995247022a423d46cd28e5fa2ec92da287a1a130ef4ca44d668fedc9d9d00da
🛡️ VirusTotal Result: ✅ Clean

🔍 Scanning File: C:\Users\VICTUS\Desktop\files\netiougc.exe
🔹 SHA-256: 8cadcf5b0df1017fed04e4c2b75e54f737a2a9cf0bedd62378b712ce69eec001
```

Before fix : Files were processed one by one → slow for large directories.
After Fix: Used concurrent.futures.ThreadPoolExecutor to process multiple files in parallel → much faster scanning.